



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN COMPUTER NETWORKS & CYBER SECURITY
COURSE OUTLINE

Unit Title and Code: CNS 2106 INTERNET PROGRAMMING

Lecturer: Lawrence Muriira, D.Eng.

Class Rep:

Staff Room: Hybrid

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Aim: To develop advanced skills in programming dynamic web applications using web technologies.

Course Learning Outcomes:

- Develop Object Oriented PHP Applications.
- Apply frameworks in web development.
- Create dynamic web applications for on-line transactions.
- Integrate web applications with current scripting technologies

Contact Hours: 45

Prerequisite:

Content: Introduction to HTML, OOP PHP; Persistent Data Objects; Introduction to PHP Frameworks such as CodeIgniter or Laravel; PHP Security; Data Formats & Types (XML Basics, XML Extension, XPath; Web Services Basics; SOAP, REST, JSON and AJAX; DATETIME and DOM Document); String Manipulation; Input/Output; Advanced PHP techniques.

Week/Dates	Topic	Intended Learning Outcomes The students will be able to:	Activities
1	Introduction	• Understand the Background & Introduction to HTML programming • Understand the Basic Terminology and Definitions • HTML Coding	• Class discussions • Lecture • Class exercise
2	Introduction to CSS/JavaScript	• Understand the Background & Introduction to server-side scripting • Understand the Basic	• Lectures • Class discussions • Class exercise • Labs Practical

		Terminology and Definitions • Server-side scripting	
3.	Introduction to PHP	• Understand the Background & Introduction to PHP programming • Understand the Basic Terminology and Definitions	• Class discussions • Lecture
4.	Introduction to PHP Environment and coding	• Explain PHP Syntax • Write PHP Comments, Variables, Echo / Print, Data Types, Strings • Apply PHP Numbers, Math, Constants, Operators • Write control processes such as PHP If...Else and Elseif, PHP Switch and PHP Loops •	• Class discussions • Lecture • Class exercise RAT
5.	Introduction to PHP coding II	• Write PHP Functions • Write PHP Arrays	• Lectures • Class discussions • Class exercise • Labs Practical Assignment 1
6.	PHP Forms	• Create PHP Forms, Form Handling, Form Validation • Enable PHP Form Required • Add PHP Form URL/E-mail Create PHP Form Complete	• Lectures • Class discussions • Class exercise • Labs Practical Assignment 2
7.	Record and File Processing	• Write PHP Date and Time • PHP Include • Create PHP File Handling, File Open/Read, File Create/Write, File Upload • Enable PHP Cookies, Sessions	• Lectures • Class discussions • Class exercise • Labs Practical Assignment 3 CAT 1
8.	PHP Object-Orientation	• Explain PHP OOP, What is OOP • Write PHP Classes/Objects, Constructor, Destructor • Create Access Modifiers, Inheritance, Constants, and Abstract Classes • Write PHP Interfaces, Traits, Static Methods, Static Properties • Create PHP Namespaces and Iterables	• Lectures • Class discussions • Class exercise • Labs Practical Assignment 4

		Group Creation and Projects	
9.	Web Database	- Explain MySQL Database - How MySQL Connects - Create MySQL DB and Create Table - Insert Data in MySQL - Get MySQL Last ID - Insert Multiple, Prepared, and Select Data in MySQL - Use MySQL Where, Order By, Delete Data, and Update Data commands - Enable MySQL Limit Data	- Lectures - Class discussions - Class exercise - Labs Practical Assignment 5
10.	Project Progress presentation	Present the progress of their projects in two days per group	Presentations
11.	XML Basics and Extentions	- Explain PHP XML Parsers - Insert Simple XML Parser, Get, Expat and DOM in PHP code.	- Lectures - Class discussions - Class exercise - Labs Practical Assignment 6
12.	Revision	Final Exam Preparation	- Class discussions - Class exercise - Labs Practical Assignment 8 CAT 2
13.	Project Presentation and marking		

Course Delivery Methodology

Lectures and presentations in class, Program simulations, practical assignments and CATs, software design competition among students, and any other interactive methodology.

Academic Assessment

Type	Weighting (%)
Examination	60
CAT1	<u>10</u>
CAT2	<u>10</u>
Assignment WEEKLY	<u>10</u>
Labs AFTER EVERY TWO WEEKS	<u>10</u>
Total	<u>100 %</u>

Course Reference Materials

Books

- Deitel, P., Deitel, H., & Abbey, D. (2012). Internet & World Wide Web. (5, Ed.) New Jersey: Prentice-Hall. (Available in the university library QA76. 625.D47 2012)
- Luke, W., & Laura, T. (2017). PHP and MySQL Web Development (5th Edition) (Developer's Library). Addison-Wesley Professional (Available in the university library QA76.73. P224 W45 2017)
- Kevin Tatroe, Peter MacIntyre and Rasmus Lerdorf . Programming PHP . (2013), 1449392776
- Williams, H. Web database applications with PHP and MySQL

Recommended Books

- Wicked cool PHP by Steinmetz, William.
- Object-oriented PHP by Lavin, Peter.

Websites

- PHP: Hypertext Preprocessor (<http://www.php.net>)
- W3Schools Online Web Tutorials (<https://www.w3schools.com/php/default.asp>)
- Tutorial Point (https://www.tutorialspoint.com/php/php_tutorial.pdf)

Classes

1. Punctuality is fundamental.
2. Active participation in class discussions is encouraged and could earn marks

Communication Channels

- *Via E-mail or*
- *Module Leader (Class Rep)*